Ultra Evaluation Function

How Laser War measures a position

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Central idea. A positive score favors the player whose turn it is. A negative score favors the opponent. Ultra values concrete route proximity and one-move pressure, then adjusts for surviving shields and a laser already contacting friendly cover.

1. Terminal positions

The game result overrides every positional estimate:

$$E = \begin{cases} +10000, & \text{the side to move is recorded as the winner,} \\ -10000, & \text{the opponent is recorded as the winner,} \\ 0, & \text{the position is a draw.} \end{cases}$$

During search, shorter wins and longer losses receive small ply adjustments, so Ultra prefers the quickest proven win and delays a proven loss.

2. Route measurements

For each of the two lasers, the rules engine computes the fewest additional mirrors needed to reach each king. Destroying a shield costs one step and placing a turning mirror costs one step. Missing or very long routes are capped at 12.

Let $a_1 \le a_2$ be the sorted costs to the opposing king and let $d_1 \le d_2$ be the sorted costs to the current player's king. Define the bounded proximity

$$p(c) = 12 - c.$$

The first cost describes the quickest available laser; the second describes the reserve laser. Keeping both prevents one attractive route from hiding complete weakness on the other side of the board.

Also define

$$A_1 = \#\{a_i \le 1\}, \qquad D_1 = \#\{d_i \le 1\}.$$

These counts identify positions in which one or both lasers need at most one new mirror to reach a king. Held-out game data found these one-move counts more useful than a large generic bonus for visually open routes.

3. Shields and live contact

Let N_o and N_e be the surviving shields around the current player's king and the opposing king. Shields never move or regenerate, so no separate shape term is used.

Let C_d be the number of current lasers whose next collision is a shield belonging to the current player. Such contact is dangerous because the shield is already scheduled to disappear on the next volley. Contact with an opposing shield was tested, but its fitted coefficient was zero after route pressure was included, so it contributes nothing here.

4. Complete non-terminal score

For every non-terminal position,

$$\begin{aligned} E = B &+ w_o N_o + w_e N_e \\ &+ w_{a1} p(a_1) + w_{d1} p(d_1) \\ &+ w_{a2} p(a_2) + w_{d2} p(d_2) \\ &+ w_A A_1 + w_D D_1 + w_C C_d. \end{aligned}$$

The signed constants below are rounded for readability; the source records their full precision. A negative danger coefficient means that an easier route to the current player's king lowers the score. The constant B calibrates the raw sum to observed outcomes; it does not represent material on the board.

Signal	Meaning	Weight
Calibration baseline B	Centers the fitted score	-1700.63
Own surviving shield w_o	Remaining friendly cover	+90.43
Opponent surviving shield w_e	Remaining enemy cover	-45.33
Nearest attack w_{a1}	Quickest route to the enemy king	+174.37
Nearest danger w_{d1}	Quickest route to the friendly king	-5.50
Reserve attack w_{a2}	Second route to the enemy king	+22.10
Reserve danger w_{d2}	Second route to the friendly king	-36.79
Attack within one move w_A	Immediately buildable attack route	+396.52
Danger within one move w_D	Immediately buildable enemy route	-164.99
Friendly shield under laser w_C	Cover already contacted by a beam	-123.57

5. Why earlier terms were removed

The previous evaluator separately rewarded permanent one-king assignment, current king exposure, nonlinear route gaps, and raw shield cascades. Those ideas are tactically meaningful, but grouped ablations found that they did not improve held-out prediction after granular route distance and one-move pressure were known. They are therefore not assigned unsupported static weights.

This does not make Ultra tactically blind. Search still treats an actual win as ± 10000, checks every legal mate in one, extends loaded king exposures, and can certify short forcing lines. Those are discrete tactical facts, not soft positional bonuses.

6. Evidence and interpretation

The fitted model was trained with monotonic signs: attack features could not become negative and danger features could not become positive. Games, not individual positions, defined the train/hold-out split.

Compared with the previous formula, the selected features improved hold-out log loss by 0.0683, with a game-bootstrap 95% interval from 0.0199 to 0.1143. On a separate seed set, log loss improved by 0.1007 and outcome accuracy rose from 66.9% to 72.0%. These measurements justify an arena candidate; paired games remain the final test of playing strength.

- Scores are comparative search values, not win probabilities.
- A score near zero means the measured factors are balanced, not that a draw is forced.
- Values near ±10000 are reserved for completed or certified wins.
- One-move route pressure is deliberately stronger than raw shield material.